scanning is used by fax machines and airport x-ray baggage scanners. As a variation, the object can be kept stationary while the detector is moved. This is very convenient when the detector is already mounted on a moving object, such as an aircraft taking images of the ground beneath it. The advantage of line scanning is that *speed* is traded for detector *simplicity*. For example, a fax machine may take several seconds to scan an entire page of text, but the resulting image contains thousands of rows and columns.

An even more simplified approach is to acquire the image **point-by-point**. For example, the microwave image of Venus was acquired one pixel at a time. Another example is the *scanning probe microscope*, capable of imaging individual atoms. A small probe, often consisting of only a single atom at its tip, is brought exceedingly close to the sample being imaged. Quantum mechanical effects can be detected between the probe and the sample, allowing the probe to be stopped an exact distance from the sample's surface. The probe is then moved over the surface of the sample, keeping a constant distance, tracing out the peaks and valleys. In the final image, each pixel's value represents the elevation of the corresponding location on the sample's surface.

Printed images are divided into two categories: **grayscale** and **halftone**. Each pixel in a grayscale image is a shade of gray between black and white, such as in a photograph. In comparison, each pixel in a halftone image is formed from many individual *dots*, with each dot being completely black or completely white. Shades of gray are produced by alternating various numbers of these black and white dots. For example, imagine a laser printer with a resolution of 600 dots-per-inch. To reproduce 256 levels of brightness between black and white, each pixel would correspond to an array of 16 by 16 printable dots. Black pixels are formed by making all of these 256 dots black. Likewise, white pixels are formed making all of these 256 dots white. Mid-gray has one-half of the dots white and one-half black. Since the individual dots are too small to be seen when viewed at a normal distance, the eye is fooled into thinking a grayscale has been formed.

Halftone images are easier for printers to handle, including photocopy machines. The disadvantage is that the image quality is often worse than grayscale pictures.

Brightness and Contrast Adjustments

An image must have the proper **brightness** and **contrast** for easy viewing. Brightness refers to the overall lightness or darkness of the image. Contrast is the *difference* in brightness between objects or regions. For example, a white rabbit running across a snowy field has *poor* contrast, while a black dog against the same white background has *good* contrast. Figure 23-10 shows four possible ways that brightness and contrast can be misadjusted. When the brightness is too high, as in (a), the whitest pixels are saturated, destroying the detail in these areas. The reverse is shown in (b), where the brightness is set too low, saturating the blackest pixels. Figure (c) shows